DATA SCIENCE V1.0

Lesson 6

Documentation: Data Science V1.0

Generative AI & RAG Systems

Objectives

- Understand how LLMs work
- Implement RAG pipeline
- Work with embeddings
- Understand hallucination risks

Topics

- Transformer architecture (conceptual)
- Prompt engineering
- Embeddings
- Vector similarity
- RAG architectures
- Chunking strategies

Resources

- OpenAI documentation: https://platform.openai.com/docs
- LangChain: https://python.langchain.com/docs/
- LlamaIndex: https://docs.llamaindex.ai/

Practice Project – Document Q&A RAG System
Requirements

- WebScraping some wikipedia pages on a specific topic/scrap a small website
- Text chunking
- Generate embeddings
- Retrieve top-k relevant chunks
- Generate response

Must Include

- Similarity score explanation
- Chunk size experimentation
- Failure case analysis
- Hallucination example